

Erdős Problem #1066

EP-1066

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let G be a graph given by n points in $\mathbb{R}^2$, where any two distinct points are at least distance 1 apart, and we draw an edge between two points if they are distance 1 apart.

Let $g(n)$ be maximal such that any such graph always has an independent set on at least $g(n)$ vertices. Estimate $g(n)$, or perhaps $\lim \frac{g(n)}{n}$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>